

An Infinite Block-Incidence Ladder and a Four-Block Obstruction for Hénon Galois Excess

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Abstract

For a certified mixing symbolic survivor of the area-preserving Hénon map, the Mahler height of each primitive return multiplier splits into physical instability length and a nonnegative Galois excess $\mathcal{E}_\gamma$. An earlier finite test ruled out potentials depending on at most three symbols but left the required all-period structure unspecified. We construct two explicit primitive families

$$A_m = 0^{m-2}21, \quad B_m = 0^{m-3}231$$

and prove, for every $m \geq 3$, the exact cyclic incidence identity

$$\mathbf{N}_m(A_m) + \mathbf{N}_m(B_{m+2}) = \mathbf{N}_m(A_{m+1}) + \mathbf{N}_m(B_{m+1}).$$

Both differences reduce to the same signed three-atom insertion row. The first new member needed to test this ladder, B_6 , has an exact radical orbit, trace $18062 + 5352\sqrt{7}$, and irreducible reciprocal degree-four multiplier polynomial. Its Galois excess is $\text{arcosh}(9031 - 2676\sqrt{7})$. Combining this with the shared period-five trace field yields a strict exact inequality opposite to the forced $m = 4$ periodic-sum identity. Hence no locally constant potential of width at most four realizes all Galois excesses. A determinant-one seven-row minor shows that the same finite witness is interpolable at width five, so the obstruction is finite-level sharp. For a one-sided Hölder realization the infinite ladder gives the explicit necessary condition $|\Delta_m| \leq C(4m + 4)\vartheta^{\alpha m}$. Determining the asymptotics of Δ_m is now the single remaining regularity gate; no general Hölder no-go or arithmetic spectral conclusion is claimed.

1 Introduction

Thermodynamic zeta functions turn periodic-orbit sums of a regular potential into an Euler product or transfer-operator determinant [1, 4]. This additive interface is essential: an arbitrary assignment of one number to every primitive orbit need not be the periodic data of any single observable.

For the certified H6 Hénon survivor, HCS-P54 decomposed the Mahler height of every algebraic return multiplier into

$$\mathcal{H}_\gamma = \ell_\gamma + \mathcal{E}_\gamma, \quad \mathcal{E}_\gamma \geq 0. \quad (1)$$

The physical roof ℓ_γ has a source-backed pressure pole, while a regular realization of the nonphysical term $\mathcal{E}_\gamma$ remains the completion gate [6]. HCS-P55 attacked its finite-memory forms. Five exact cycles gave a triple-block incidence relation whose corresponding excess identity failed, proving that width at most three is impossible [7]. More importantly, that paper showed why isolated

finite counterexamples cannot settle unrestricted Hölder realizability: any finite orbit list can be interpolated on sufficiently long cylinders.

The present paper makes the next step nonlocal in the block width. We find two natural homoclinic-looking cycle families, prove one incidence relation for every width, and calculate the first new exact orbit required by the family. The result has three layers.

First, the symbolic relation is a theorem for all $m \geq 3$, not a computer pattern. Increasing either family by one period inserts one zero into its long zero run. At width m , both insertions change the incidence vector by exactly the same signed three-atom row. Subtraction produces the infinite ladder.

Second, the first untested ladder equation, at $m = 4$, fails by a wide exact margin. The new period-six orbit has a quadratic trace field and a degree-four multiplier field. A single nonphysical period-five embedding appears in the excess of each of the two period-five cycles, giving a clean lower bound. Elementary square and logarithm inequalities then prove the opposite of the forced identity without relying on decimal separation.

Third, a unimodular width-five minor interpolates all seven finite witness rows. Thus the conclusion is exactly scoped: no width-at-most-four local potential exists, but these seven rows do not obstruct width five. The infinite ladder instead supplies the concrete Hölder target Δ_m : every one-sided Hölder realization forces exponential decay of this explicit sequence. Establishing or refuting that decay is a genuine all-period theorem, not another finite-rank calculation.

No rational-prime orbit law, completed determinant, trace formula or Hilbert–Pólya operator is constructed here. The advance is a sharper regularity obstruction and an exact reusable ladder inside Route A.

2 Frozen survivor and cyclic incidence

We retain without refitting the area-preserving map

$$H_6(q, p) = (1 - 6q^2 - p, q), \quad \det DH_6 = 1, \quad (2)$$

and the certified mixing four-state survivor from HCS-P31 [2, 5]. Its labels and adjacency matrix are

$$0 = --, \quad 1 = -+, \quad 2 = +-, \quad 3 = ++, \quad A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

For a cyclic word $\gamma = (x_0, \dots, x_{n-1})$, let

$$\mathbf{N}_r(\gamma; w) = \#\{0 \leq j < n : (x_j, \dots, x_{j+r-1}) = w\}, \quad (3)$$

with indices modulo n . A width- r potential $\varphi(x) = v(x_0, \dots, x_{r-1})$ has periodic sum

$$\mathbf{S}_\gamma \varphi = \sum_w v(w) \mathbf{N}_r(\gamma; w). \quad (4)$$

Consequently every linear relation among cyclic incidence rows forces the same relation among the periodic sums of every width- r potential.

For a primitive orbit, let $M_\gamma \in \mathrm{SL}_2$ be its chronological monodromy and $\Lambda_\gamma > 1$ its physical expanding modulus. If f_γ is the monic minimal polynomial of the signed multiplier, then

$$\mathcal{H}_\gamma = \log M(f_\gamma), \quad \ell_\gamma = \log \Lambda_\gamma, \quad \mathcal{E}_\gamma = \mathcal{H}_\gamma - \ell_\gamma. \quad (5)$$

Reciprocity pairs the roots of f_γ ; all expanding nonphysical pairs contribute nonnegative terms to $\mathcal{E}_\gamma$, as certified in HCS-P54.

3 An infinite incidence ladder

Define two admissible cyclic families

$$A_m = 0^{m-2}21 \quad (m \geq 3), \quad B_m = 0^{m-3}231 \quad (m \geq 4). \quad (6)$$

Every displayed transition is allowed by A , including the cyclic return $1 \rightarrow 0$. Each word contains exactly one symbol 2, so it cannot be a proper power. Thus all members are primitive.

Write e_w for the coordinate vector of a block w .

Lemma 3.1 (common insertion row). *For every $m \geq 3$,*

$$\mathbf{N}_m(A_{m+1}) - \mathbf{N}_m(A_m) = e_{0^{m-1}2} + e_{10^{m-1}} - e_{10^{m-2}2}, \quad (7)$$

$$\mathbf{N}_m(B_{m+2}) - \mathbf{N}_m(B_{m+1}) = e_{0^{m-1}2} + e_{10^{m-1}} - e_{10^{m-2}2}. \quad (8)$$

Proof. In each comparison one extra zero is inserted at the beginning of the unique maximal zero run. List the cyclic width- m windows by their start relative to the unique terminal symbol 1. Windows that see the same part of the zero run cancel bijectively. The longer word adds the windows $0^{m-1}2$ and 10^{m-1} , while the shorter word has the unmatched window $10^{m-2}2$. The symbols after 2 distinguish the two families but lie in the portion common to both sides of each comparison, so the same three survivors remain. This gives both identities. $\square$

Theorem 3.2 (all-width block-incidence ladder). *For every integer $m \geq 3$,*

$$\mathbf{N}_m(A_m) + \mathbf{N}_m(B_{m+2}) = \mathbf{N}_m(A_{m+1}) + \mathbf{N}_m(B_{m+1}). \quad (9)$$

Proof. Subtract (7) from (8) and rearrange. $\square$

The cases $m = 3$ and $m = 4$ recover respectively the P55 relation and its first extension:

$$\begin{aligned} \mathbf{N}_3(A_3) + \mathbf{N}_3(B_5) &= \mathbf{N}_3(A_4) + \mathbf{N}_3(B_4), \\ \mathbf{N}_4(A_4) + \mathbf{N}_4(B_6) &= \mathbf{N}_4(A_5) + \mathbf{N}_4(B_5). \end{aligned}$$

The exact program checks the formula independently through $m = 64$, but that finite check is evidence for the implementation, not the proof.

4 Exact algebra of the new period-six orbit

The new word $B_6 = 000231$ has sign coordinates $---++-$. Set

$$c = -\frac{\sqrt{7}}{6}, \quad D = \sqrt{25 + 4\sqrt{7}}, \quad a = \frac{-1 - D}{12}, \quad d = \frac{-1 + D}{12}. \quad (10)$$

Proposition 4.1 (radical period-six certificate). *The cyclic coordinate tuple*

$$(q_0, \dots, q_5) = (a, a, c, d, d, c)$$

satisfies $q_{j+1} = 1 - 6q_j^2 - q_{j-1}$ exactly and has sign word $---++-$. Its monodromy trace and trace polynomial are

$$T_6 = 18062 + 5352\sqrt{7}, \quad (11)$$

$$Q_6(T) = T^2 - 36124T + 125728516. \quad (12)$$

The signed unstable multiplier has irreducible reciprocal polynomial

$$z^4 - 36124z^3 + 125728518z^2 - 36124z + 1. \quad (13)$$

Its Galois excess is

$$\mathcal{E}_{B_6} = \operatorname{arcosh}(9031 - 2676\sqrt{7}). \quad (14)$$

Proof. Substitution of (10) into the six cyclic recurrence equations gives zero. Multiplying the derivative matrices

$$DH_6(q_j, q_{j-1}) = \begin{pmatrix} -12q_j & -1 \\ 1 & 0 \end{pmatrix}$$

gives determinant one and trace (11). Conjugating $\sqrt{7} \mapsto -\sqrt{7}$ yields (12). Eliminating $T = z + z^{-1}$ gives (13); direct factorization over $\mathbb{Q}$ is irreducible. Concretely, modulo 13 it becomes

$$\bar{f}(z) = z^4 + 3z^3 + 6z^2 + 3z + 1,$$

and $\gcd(\bar{f}, z^{13} - z) = \gcd(\bar{f}, z^{169} - z) = 1$. Thus it has no linear or quadratic factor over $\mathbb{F}_{13}$, proving irreducibility over $\mathbb{Q}$. The nonphysical trace is $18062 - 5352\sqrt{7}$, so its expanding reciprocal pair contributes $\operatorname{arcosh}(|T'|/2)$, which is (14). $\square$

The conjugate half-trace is isolated without floating point:

$$1950 < 9031 - 2676\sqrt{7} < 1951. \quad (15)$$

Indeed

$$7081^2 - 7 \cdot 2676^2 = 13729 > 0, \quad 7 \cdot 2676^2 - 7080^2 = 432 > 0.$$

Numerically, $\mathcal{E}_{B_6} = 8.26922881806116\dots$; decimals are not used in the obstruction proof.

5 The four-block obstruction

The two period-five words $A_5 = 00021$ and $B_5 = 00231$ belong to the same degree-six trace field certified in HCS-P55. Its polynomial is

$$\begin{aligned} Q_5(T) = & T^6 + 3300T^5 - 34165368T^4 - 7291075328T^3 \\ & + 26529205510272T^2 + 3609165326736384T - 4266315336505009664. \end{aligned} \quad (16)$$

It has exactly one root in each interval

$$(-7607, -7606), (-711, -710), (-590, -589), (390, 391), (770, 771), (4445, 4446). \quad (17)$$

The physical roots for A_5 and B_5 lie in the first and last intervals respectively. Therefore the root in $(-711, -710)$ is nonphysical for both, and

$$\mathcal{E}_{A_5} > \operatorname{arcosh} 355, \quad \mathcal{E}_{B_5} > \operatorname{arcosh} 355. \quad (18)$$

For $A_4 = 0021$, the only nonphysical half-trace is $287 - 96\sqrt{6} \in (51, 52)$, whence

$$\mathcal{E}_{A_4} < \operatorname{arcosh} 52. \quad (19)$$

Equation (15) gives

$$\mathcal{E}_{B_6} < \operatorname{arcosh} 1951. \quad (20)$$

Theorem 5.1 (four-block no-go). *No locally constant potential depending on at most four consecutive $H6$ states has periodic sums $\mathcal{E}_\gamma$ on every primitive orbit.*

Proof. For $x > 1$,

$$\log(2x - 1) < \operatorname{arcosh} x < \log(2x),$$

because $x - 1 < \sqrt{x^2 - 1} < x$. Moreover

$$709^2 = 502681 > 405808 = 104 \cdot 3902.$$

Consequently

$$2 \operatorname{arcosh} 355 > 2 \log 709 > \log 104 + \log 3902 > \operatorname{arcosh} 52 + \operatorname{arcosh} 1951.$$

Combining this with (18)–(20) proves

$$\mathcal{E}_{A_5} + \mathcal{E}_{B_5} > \mathcal{E}_{A_4} + \mathcal{E}_{B_6}. \quad (21)$$

But Theorem 3.2 at $m = 4$ forces equality for every width-four potential. A smaller-width potential is also a width-four potential after ignoring the extra coordinates, so every width at most four is excluded. $\square$

For scale only, the signed ladder discrepancy with left-minus-right convention is

$$\mathcal{E}_{A_4} + \mathcal{E}_{B_6} - \mathcal{E}_{A_5} - \mathcal{E}_{B_5} = -55.5475038022260 \dots$$

The theorem rests on the exact logarithmic chain, not this decimal.

6 Finite sharpness at width five

The four-block relation is not a general finite-data obstruction in disguise. Consider the seven rows

$$C_1 = (0), \ A_3, \ A_4, \ B_4, \ A_5, \ B_5, \ B_6$$

and the seven width-five blocks

$$00000, \ 00021, \ 00023, \ 00210, \ 00231, \ 02102, \ 02310. \quad (22)$$

Their incidence minor, in the displayed row and column order, is

$$M_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}, \quad \det M_5 = 1. \quad (23)$$

Proposition 6.1 (seven-row finite sharpness). *Every prescribed real total on these seven cycles is realized by a width-five locally constant potential. In particular, the Galois-excess witness is sharp at width five.*

Proof. The determinant in (23) is one. More explicitly, for totals $(E_1, E_3, E_{4A}, E_{4B}, E_{5A}, E_{5B}, E_6)$, assign to the blocks in (22) the values

$$E_1, -E_{4A} + E_{5A}, -E_{5B} + E_6, E_{4A}, -E_{4B} + E_{5B}, E_3, E_{4B},$$

and assign zero to all other width-five blocks. Multiplication by M_5 returns the desired totals. $\square$

The coefficients need not be nonnegative; the realization problem is for a real potential. Proposition 6.1 forbids an overstatement: Theorem 5.1 does not itself disprove arbitrary finite-memory or Hölder realizations. It locates the first four-block failure and identifies the all-width sequence that must be studied next.

7 The explicit Hölder discrepancy sequence

Let Σ_A^+ carry the metric $d_\vartheta(x, y) = \vartheta^{N_+(x, y)}$, where $0 < \vartheta < 1$ and N_+ is the common-prefix length. If φ is α -Hölder, approximation by a width- m cylinder function has uniform error at most $C\vartheta^{\alpha m}$. Applying the incidence identity (9) to that approximation yields the following specialization of the quantitative gate from HCS-P55.

Corollary 7.1 (ladder condition for one-sided Hölder data). *If one α -Hölder potential on Σ_A^+ satisfies $S_\gamma\varphi = \mathcal{E}_\gamma$ for every primitive cycle, then for all $m \geq 3$,*

$$\Delta_m := \mathcal{E}_{A_m} + \mathcal{E}_{B_{m+2}} - \mathcal{E}_{A_{m+1}} - \mathcal{E}_{B_{m+1}} \quad (24)$$

satisfies

$$|\Delta_m| \leq C(4m + 4)\vartheta^{\alpha m}. \quad (25)$$

Proof. The locally constant approximation is killed exactly by (9). The four orbit lengths are $m, m + 2, m + 1, m + 1$, whose sum is $4m + 4$. Bounding the approximation error on each orbit gives (25). $\square$

This is the central upgrade over a vague search for “more relations.” One fixed, closed-form family now tests every forward scale. To refute a one-sided Hölder realization it would suffice to prove that Δ_m fails every bound of the form $Cm\rho^m$ with $0 < \rho < 1$. Conversely, observed decay is not by itself a construction: one would still need a compatible extension producing a single potential on the whole subshift.

No two-sided conclusion is automatic. Forward block counts control a future-dependent representative. Applying the same condition to an arbitrary two-sided Hölder potential requires an explicit cohomological reduction to such a representative, in the spirit of Livšic theory [3]; periodic data alone do not supply that reduction.

The exact data presently give $\Delta_3 > 0$ and $\Delta_4 < 0$, both large in magnitude. Two values do not determine asymptotics. The next major theorem is therefore a recurrence, renormalization law, or certified asymptotic estimate for the Galois excesses along A_m and B_m .

8 Route status, reproducibility and conclusion

The primary exact producer locks seven upstream artifacts, enumerates all primitive symbolic cycles through period six, verifies the insertion formula through width 64, derives the period-six field, isolates the relevant period-five roots, checks the logarithmic obstruction and rejects twenty claim

mutations. An independent program uses DFS and **Counter** incidence rows, recomputes the exact algebra and compares final invariants. Fifteen unit tests exercise the ladder, field, obstruction, interpolation and scope firewalls. Run the full certificate from the project directory with

```
bash code/run_c56.sh
```

No prime table, zero table, fitted potential or post-hoc orbit choice enters the computation.

Under the Route-A evaluator the status remains

```
A1 = A1_WEAK,
A2 = A2_ANALYTIC_DETERMINANT (physical subsystem only),
A3 = A3_PARTIAL_ANALYTIC_STRUCTURE,
A4 = A4_FORMAL_HINT,
```

with overall **ROUTE_A_EXPLORATORY**. The all-width ladder is a stronger structural diagnosis, but it does not give the full Galois-weighted candidate an analytic determinant. Route B is not authorized: no Hilbert space, operator domain, self-adjointness theorem, target spectral type, von-Mangoldt trace, or completed determinant is provided.

The strongest positive result is the infinite incidence ladder together with the exact degree-four multiplier field of B_6 . The strongest obstruction is the strict failure of its $m = 4$ excess identity. The reusable structure is the common three-atom insertion row. The open theorem is the asymptotic behavior of Δ_m , and the most direct next experiment is exact continuation to $m = 5, 6, \dots$ coupled to a search for a recurrence in the trace or excess fields.

Accordingly this paper advances the local regularity bridge but not the arithmetic bridge. It claims neither a rational-prime correspondence nor a Riemann-zero spectral realization.

A Exact and numerical ledger

For reproducibility, the principal certified values are

quantity	exact value or interval	decimal value
$\text{tr } M_{B_6}$	$18062 + 5352\sqrt{7}$	32222.0610...
$\mathcal{E}_{B_6}$	$\text{arcosh}(9031 - 2676\sqrt{7})$	8.2692288181...
$\mathcal{E}_{A_4}$	$\text{arcosh}(287 - 96\sqrt{6})$	4.6413895255...
$\mathcal{E}_{A_5}$	period-five height minus physical root	33.9605061148...
$\mathcal{E}_{B_5}$	period-five height minus physical root	34.4976160310...
Δ_4	certified strictly negative	-55.5475038022...

The exact strictness margin in the final logarithm comparison is

$$709^2 - 104 \cdot 3902 = 96873.$$

The seven-by-seven width-five minor has determinant one. The machine-readable certificate records every matrix row, all exact polynomial coefficients, the 20 rejected mutations and the SHA-256 locks of all inherited artifacts.

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